

Equity US — Grinold-Kroner Model

Complete Assumption Relationship Schematic

MACRO: US BUILDING BLOCKS

GDP Building Blocks

population_growth 0.40%
productivity_growth 1.20%
my_ratio 2.1
-> sigmoid --0.10%
adjustment -0.30%

Inflation Building Blocks

current_headline_infl 2.50%
weight: 30%
long_term_inflation 2.20%
weight: 70%

T-Bill Building Blocks

current_tbill 3.67%
weight: 30%
country_factor 0.00%
long_term = GDP + Infl + CF
weight: 70%

E[Real GDP] = 1.20%

= (1.20 + sigmoid + adj) + 0.40
= 1.20%

E[Inflation] = 2.29%

= 30% x 2.50 + 70% x 2.20
= 2.29%

E[T-Bill] = 3.54%

= 30% x 3.67 + 70% x (0+1.20+2.29)
= 3.54%

Direct Forecast Overrides

User can override GDP / Inflation directly to BREAK the building-block link

No direct link
to equities

(feeds Liquidity, Bonds,
Hedge Funds instead)

macro_rgdp
= 1.20%

macro_inflation
= 2.29%

EQUITY US — GRINOLD-KRONER COMPONENTS

1. Dividend Yield

dividend_yield = 1.30% [INPUT]
Current trailing 12-month yield, no mean reversion

2. Net Buyback Yield

net_buyback_yield = 1.50% [INPUT]
Gross buybacks minus dilution from issuance

3. Revenue Growth <-- MACRO-LINKED (key relationship)

DEFAULT (auto-computed from macro):

revenue_growth = macro_inflation + macro_rgdp + revenue_gdp_wedge
5.49% = 2.29% + 1.20% + 2.00%

If user overrides revenue_growth directly, the macro link is BROKEN

4. Margin Change

margin_change = -0.50% [INPUT]
Negative = margin compression from peak

5. Valuation Change (P/E)

current_pe = 22.0x target_pe = 20.0x [INPUT]
val = (20/22)^(1/10) - 1 = -0.95% [COMPUTED]

inflation
nominal
return

FORMULA

E[Nominal] = DY + Buyback + RevGrowth + Margin + Valuation
= 1.30 + 1.50 + 5.49 + (-0.50) + (-0.95)

OUTPUT

E[Nominal Return] = 6.84%

E[Real Return] = Nominal - Inflation
= 6.84% - 2.29%

E[Real Return] = 4.55%

macro_inflation used again here for real return back-computation

Legend:

GDP path

Macro-linked component

Inflation path

Direct equity input

T-Bill path (no equity link)

Valuation (computed)